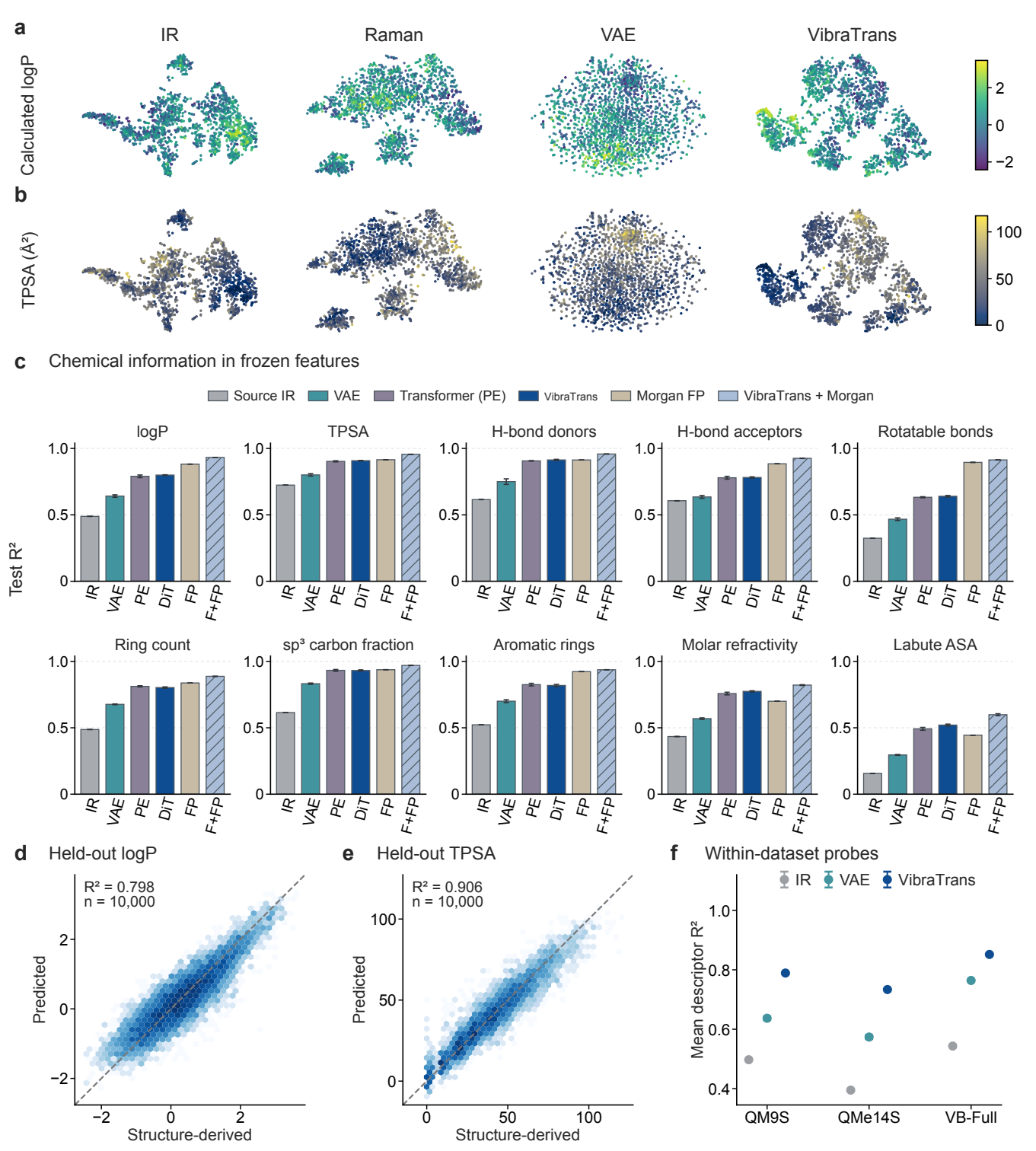

a
IR
Raman
VAE
VibraTrans
Calculated logP
2
0
−2
b
TPSA (Å²)
100
50
0
c
Chemical information in frozen features
Source IR
VAE
Transformer (PE)
VibraTrans
Morgan FP
VibraTrans + Morgan
Test R²
logP
TPSA
H-bond donors
H-bond acceptors
Rotatable bonds
Ring count
sp³ carbon fraction
Aromatic rings
Molar refractivity
Labute ASA
IR
VAE
PE
DiT
FP
F+FP
d
Held-out logP
R² = 0.798
n = 10,000
Predicted
Structure-derived
e
Held-out TPSA
R² = 0.906
n = 10,000
Predicted
Structure-derived
f
Within-dataset probes
IR
VAE
VibraTrans
Mean descriptor R²
QM9S
QMe14S
VB-Full